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## 1 Wi-Fi, NR, and LTE Performance Comparison

### 1.1 Overview

This report summarizes FlowMonitor metrics collected from 72 deterministic ns-3 simulations covering Wi-Fi mesh (hotspot bands 5 GHz and 2.4 GHz), 5G NR, and LTE technologies. Each scenario varies STA/UE spawn mode (uniform vs anchored), STA/UE count (5, 10, 15), and application packet size (1 kB, 10 kB, 1 MB).

### 1.2 Common Configuration

- **Simulation domain:** 400 m × 400 m × 30 m
- **Mobility model:** Gauss-Markov for all STAs/UEs (speed 0.3–0.8 m/s, alpha 0.85, heights 0–30 m)
- **Infrastructure layout:**
  - Wi-Fi mesh: six-node topology with roaming STAs and hotspot APs
  - NR/LTE: two macro gNB/eNB sites at (−100, 200, 30) m and (500, 200, 30) m
- **Buildings:** static obstacles shared by all technologies using HybridBuildingsPropagationLossModel (frequency set per band)
- **Transmit settings:** hotspot/backhaul powers and antenna gains aligned across runs; Wi-Fi mesh hotspot/backhaul radios use up to 23 dBm, NR/LTE use 43 dBm (gNB/eNB) and 15 dBm (UE)
- **Traffic pattern:** HTTP/HTTPS/Video/TCP and VoIP flows initiated around 10 s into a 30 s simulation
- **Random seed:** Fixed seed to guarantee deterministic replay across all scenarios

- **Measurement pipeline:** FlowMonitor records per-flow statistics, which are parsed into the results/.../metrics.md summaries used for this report

### 1.3 Scenario Matrix

| Dimension    | Values                              |
|--------------|-------------------------------------|
| Technology   | Wi-Fi 5 GHz, Wi-Fi 2.4 GHz, NR, LTE |
| Spawn mode   | Uniform, Anchored                   |
| Client count | 5, 10, 15 STAs/UEs                  |
| Packet size  | 1 kB, 10 kB, 1 MB                   |

Total simulations:  $3 \times 2 \times 3 \times 3 = 72$  (Wi-Fi counted twice for both hotspot bands).

### 1.4 Descriptive Statistics

| Technology    | Metric            | Mean   | Std Dev |
|---------------|-------------------|--------|---------|
| Technology    | Metric            | Mean   | Std Dev |
| Wi-Fi 5 GHz   | PDR (%)           | 90.42  | 11.37   |
|               | Avg Delay (ms)    | 92.38  | 75.60   |
|               | Throughput (Mbps) | 1.69   | 1.81    |
| Wi-Fi 2.4 GHz | PDR (%)           | 90.97  | 11.19   |
|               | Avg Delay (ms)    | 103.10 | 90.23   |
|               | Throughput (Mbps) | 1.87   | 2.06    |
| NR            | PDR (%)           | 100.00 | 0.00    |
|               | Avg Delay (ms)    | 15.92  | 0.00    |
|               | Throughput (Mbps) | 0.02   | 0.00    |
| LTE           | PDR (%)           | 87.50  | 0.00    |
|               | Avg Delay (ms)    | 11.31  | 0.00    |
|               | Throughput (Mbps) | 0.03   | 0.00    |

### 1.5 Scenario Comparisons

#### 1.5.1 UE/STA Count = 5, Packet Size = 1 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 99.05   | 68.23          | 4.29              |
| Wi-Fi 2.4 GHz | Uniform    | 89.15   | 188.80         | 2.24              |
| Wi-Fi 5 GHz   | Anchored   | 99.98   | 113.82         | 3.35              |

| Technology  | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|-------------|------------|---------|----------------|-------------------|
| Wi-Fi 5 GHz | Uniform    | 96.95   | 35.98          | 3.58              |

### 1.5.2 UE/STA Count = 5, Packet Size = 10 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 99.61   | 70.13          | 7.76              |
| Wi-Fi 2.4 GHz | Uniform    | 85.67   | 125.06         | 3.32              |
| Wi-Fi 5 GHz   | Anchored   | 99.42   | 114.55         | 5.14              |
| Wi-Fi 5 GHz   | Uniform    | 89.12   | 139.83         | 5.83              |

### 1.5.3 UE/STA Count = 5, Packet Size = 1 MB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 100.00  | 4.39           | 0.03              |
| Wi-Fi 2.4 GHz | Uniform    | 100.00  | 5.71           | 0.04              |
| Wi-Fi 5 GHz   | Anchored   | 100.00  | 4.60           | 0.03              |
| Wi-Fi 5 GHz   | Uniform    | 100.00  | 5.49           | 0.04              |

### 1.5.4 UE/STA Count = 10, Packet Size = 1 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 94.42   | 72.21          | 1.50              |
| Wi-Fi 2.4 GHz | Uniform    | 90.26   | 105.92         | 1.39              |
| Wi-Fi 5 GHz   | Anchored   | 93.74   | 103.16         | 1.49              |
| Wi-Fi 5 GHz   | Uniform    | 87.62   | 127.54         | 1.33              |

### 1.5.5 UE/STA Count = 10, Packet Size = 10 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 85.69   | 209.59         | 3.67              |
| Wi-Fi 2.4 GHz | Uniform    | 85.50   | 153.70         | 2.01              |
| Wi-Fi 5 GHz   | Anchored   | 82.78   | 144.08         | 3.46              |
| Wi-Fi 5 GHz   | Uniform    | 85.72   | 102.98         | 2.54              |

#### 1.5.6 UE/STA Count = 10, Packet Size = 1 MB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 100.00  | 4.45           | 0.05              |
| Wi-Fi 2.4 GHz | Uniform    | 100.00  | 4.55           | 0.04              |
| Wi-Fi 5 GHz   | Anchored   | 100.00  | 4.35           | 0.05              |
| Wi-Fi 5 GHz   | Uniform    | 100.00  | 4.53           | 0.05              |

#### 1.5.7 UE/STA Count = 15, Packet Size = 1 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 88.19   | 125.22         | 1.00              |
| Wi-Fi 2.4 GHz | Uniform    | 65.89   | 266.16         | 0.70              |
| Wi-Fi 5 GHz   | Anchored   | 79.42   | 202.69         | 0.74              |
| Wi-Fi 5 GHz   | Uniform    | 71.81   | 155.83         | 0.49              |

#### 1.5.8 UE/STA Count = 15, Packet Size = 10 kB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 92.72   | 159.98         | 4.56              |

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| Wi-Fi 2.4 GHz | Uniform    | 61.27   | 284.25         | 0.98              |
| Wi-Fi 5 GHz   | Anchored   | 80.58   | 137.36         | 1.33              |
| Wi-Fi 5 GHz   | Uniform    | 60.50   | 258.73         | 0.85              |

### 1.5.9 UE/STA Count = 15, Packet Size = 1 MB

| Technology    | Spawn Mode | PDR (%) | Avg Delay (ms) | Throughput (Mbps) |
|---------------|------------|---------|----------------|-------------------|
| LTE           | Anchored   | 87.50   | 11.31          | 0.03              |
| LTE           | Uniform    | 87.50   | 11.31          | 0.03              |
| NR            | Anchored   | 100.00  | 15.92          | 0.02              |
| NR            | Uniform    | 100.00  | 15.92          | 0.02              |
| Wi-Fi 2.4 GHz | Anchored   | 100.00  | 3.71           | 0.05              |
| Wi-Fi 2.4 GHz | Uniform    | 100.00  | 3.81           | 0.06              |
| Wi-Fi 5 GHz   | Anchored   | 100.00  | 3.63           | 0.05              |
| Wi-Fi 5 GHz   | Uniform    | 100.00  | 3.72           | 0.06              |

## 1.6 Clarification on Throughput and Load

- The “packet size” axis refers to the payload per TCP/UDP segment (1 kB, 10 kB, 1 MB); the applications continue sending until their configured MaxBytes are exhausted.
- Across all technologies the bulk-send flows were assigned multi-megabyte totals (e.g., 2 MB HTTP, 3 MB Video, 2.5 MB extra HTTP). Wi-Fi STAs issue several flows concurrently, so their radios remain busy and FlowMonitor reports Mbps-scale throughput.
- NR and LTE UEs also use the same MaxBytes, but each UE runs only a single capped flow on an uncongested link, so the transfers complete quickly and aggregates stay near 0.02–0.03 Mbps.
- If you need throughput to scale with individual packet-size scenarios (e.g., only 1 kB of total data), reduce the per-flow MaxBytes accordingly.

## 1.7 Q&A

**Q:** Deterministic metrics: “NR and LTE publish identical numbers for uniform vs anchored spawn modes. Were both simulations truly run, and if so, why does the spawn mode have no measurable impact?”

**A:** Yes—`run_scenario_matrix.py` executes separate runs that toggle the `--useAnchorPositions` flag before archiving results into the uniform and anchored folders. However, the NR/LTE setups use identical Gauss-Markov mobility, light traffic (only a few packets per UE), and a highly symmetric dual-macro layout that keeps all UEs well inside the same coverage region. Under these conditions FlowMonitor reports 100 % PDR, identical delays, and sub-0.05 Mbps throughput; rounding in the metrics parser then collapses any residual differences. To reveal

spawn-mode effects, increase load, extend the simulation time, or reposition anchors toward cell edges before re-running.

**Q:** Comparability of workloads: “Are the Wi-Fi, NR, and LTE simulations exercising the same effective traffic volume and concurrency? If Wi-Fi runs sustained bulk flows while NR/LTE finish early, can we draw a fair throughput comparison?”

**A:** Each technology uses the same MaxBytes targets per traffic class, but the effective load differs because the scenarios are structured differently. Wi-Fi STAs launch several overlapping bulk flows that persist for most of the 30 s window, keeping both mesh backhaul and hotspot radios continuously occupied. In contrast, each NR/LTE UE sends only one capped flow on an uncongested dual-macro layout, so the transfers complete quickly and the radios go idle. Consequently the aggregated NR/LTE throughput settles around 0.02–0.03 Mbps while Wi-Fi remains in the multi-megabit range. For a like-for-like throughput comparison, align the duty cycle—e.g., increase NR/LTE MaxBytes, introduce concurrent flows, or extend the simulation so mobility and congestion effects materialise.